E-7 Electrolyte: Technical Note

Abstract. E-7 is a carbonate electrolyte with a borate additive developed to replace the E-5 baseline at elevated temperature. This note summarises its formulation and the first 600 cycles at 45 °C.

1. Motivation. The E-5 baseline decomposed at 45 °C and cell AUR-C103 failed at cycle 412. A solvent blend with a higher boiling point and a film-forming additive was needed.

2. Formulation. The solvent blend replaces DEC with PC and DMC. The additive NaDFOB forms a stable interphase on hard carbon.

3. Results. Cells AUR-C105 and AUR-C106 retained 89.6% and 90.2% of their capacity after 600 cycles at 45 °C, with 2.1% and 2.4% swelling.

4. Conclusion. E-7 meets the 5% swelling limit and is the candidate for milestone M2. Qualification continues to 1500 cycles under Marco Lindqvist.

Component	Role	Amount
NaPF6	Salt	1.0 M
EC	Solvent	30 vol%
PC	Solvent	30 vol%
DMC	Co-solvent	40 vol%
NaDFOB	SEI additive	1 wt%